



Accessly

Kraków without barriers

The city. Accessible for you.

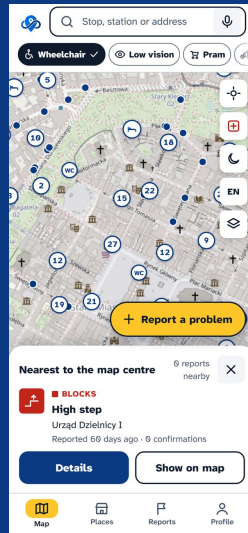
Team Accessly · Team ID: ...

HackYeah 2026 · “Cracow without barriers” challenge

Demo: <https://accessly.example.pl>

Code: <https://github.com/HakerSII/accessly>

Video (3 min): <https://youtu.be/...>



Reliable information instead of guesswork

Problem

- Accessibility data is scattered, incomplete and quickly goes out of date.
- Without a source and a date, users cannot tell whether to trust it.
- Keeping a complete database for a whole city up to date by hand does not scale.

Who it is for

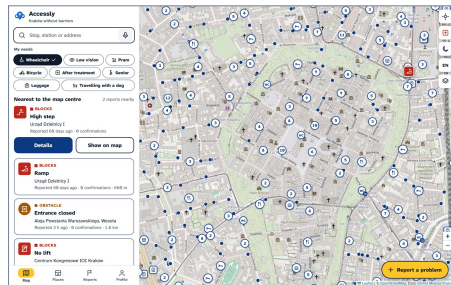
People with access needs and their carers, parents with prams, older people, people recovering from medical procedures, tourists with luggage or pets, and the medical tourism industry.

Accessly

Combines public data, information from venue owners and confirmations from the community. Every piece of information has a source, a date and a verification status.

A working prototype: a web app for phone and desktop

- **6,545 places, 27 accessibility features**
Concrete information about the entrance, lift, toilet, parking and other needs.
- **Matched to the user**
The needs profile filters and ranks places by what really matters.
- **Live, trustworthy data**
Source, date and verification status for every piece of information, plus community confirmations.
- **AI + barrier map**
The assistant recommends places, and users report and update real obstacles.



“Wheelchair” profile: places, layers, reported barriers with severity and confirmations. The list next to the map is its text alternative.

6,545

places from OSM

11

data layers

503

tests, offline

Demo: Karolina plans a trip to a café

Karolina is planning to go to a café with someone close to her who uses a wheelchair.

1 She sets the needs

Wheelchair + step-free entrance + accessible toilet

2 Accessly suggests places

A ranking of cafés matched to the profile, showing:

- what is accessible,
- what is missing,
- how up to date the data is.

3 She checks the details

Next to every fact she sees: source · date · verification status

4 She decides

She picks a place she can trust – before they leave home.

Accessly helps carers plan an outing without guesswork or stress on arrival.

Meets 0 of your 3 needs

Does not meet: Step-free entrance, Accessible toilet

No data: Wide doors (at least 90 cm). If you know, add it below or ask the owner.

Step 2: what the place meets, what is missing, what is unknown.

 **Step-free entrance** · important to you **No**

From an official source

Source: OpenStreetMap

02/10/2026

częściowo dostępne (wheelchair=limited)

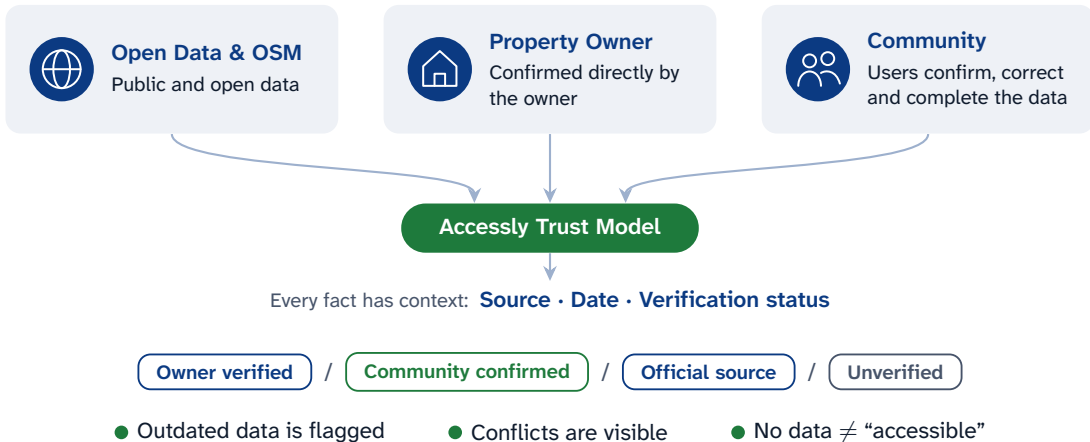
Up to date

Outdated

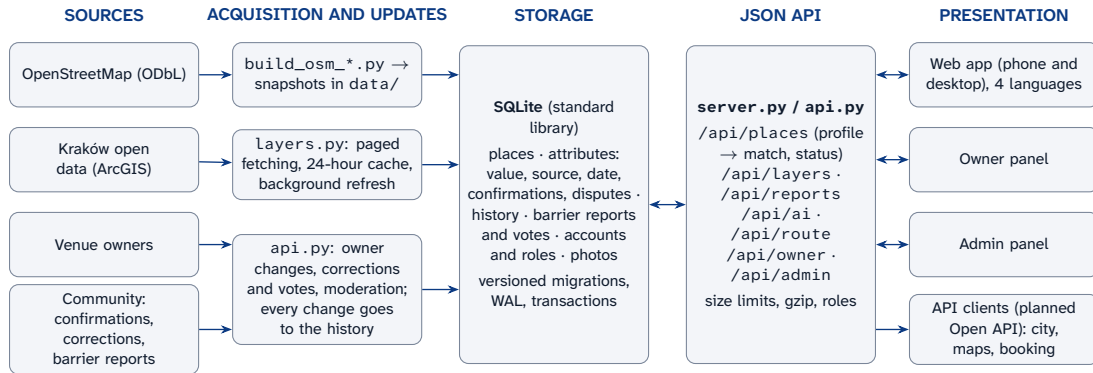
Correct

Step 3: value, source (OpenStreetMap), date (2 Oct 2026) and verification status.

Data you can trust



Architecture: data acquisition separated from presentation



Optional services, each with a fallback: Claude API (recommendations; keyword rules without a key) · OpenRouteService (wheelchair routes) · Nominatim (search from the browser, only on submit) · Rampa backend (barrier reports through the api.js adapter)

Accessible from day one

Goal: WCAG 2.2 AA

Already working



Keyboard navigation



Screen readers



Text alternative
to the map



Contrast and
legibility



Large touch targets
+ 400 % zoom



4 languages

Next step

- User testing
- NVDA / VoiceOver audit
- Full compliance of the Owner / Admin panels

Accessibility is not an add-on – it is part of Accessly's architecture.

Privacy and simple operations

Privacy by design



Use without an account, anonymous reports



Needs profile without medical data or diagnoses



Secure passwordless sign-in



HTTPS, roles and access control

Easy to deploy and run



Runs outside City Hall infrastructure



Automatic data updates



Docker – deploy on any VPS



Hosting: approx. PLN 60–120 / month



Moderation in the admin panel

Minimum user data. Minimum running costs. Maximum portability.

Business model and commercialisation

Segment	Offer	Revenue
Residents and tourists	the app, needs profile, assistant, barrier map	free, always – they create the data
Venue owners (hotels, food & drink, culture, chains)	basic panel free; Pro plan : owner badge, statistics on “what guests look for and what is missing”, bulk facilities, an accessibility widget for the website and the booking system	subscription per venue or a package for chains
Event organisers, property managers	time-limited package: temporary entrances and routes, a barrier map for the event	fee per event / venue
Cities and regions	white-label deployment (configuration, OSM import, city layers); reports on gaps in data and facilities	deployment + annual maintenance
Maps, travel apps, booking systems	Open API with attribution: accessibility facts with source, date and status	API licence / request limits

Principles that protect the value of the data. We do not sell personal data; community content and OSM data stay open (ODbL). Owners pay for tools and visibility, not for a “better” status – the status always follows verification.

From prototype to service

What we have today

- A prototype for phone and desktop: 6,545 places, 27 attributes, 11 layers, barrier map, assistant, 4 languages.
- An owner panel and an admin panel (queue, statistics, data gaps).
- Source, date and status on every fact; missing data shown as missing.
- Checked WCAG 2.2 A/AA criteria on the main scenario; 503 tests, Docker, Ansible, HTTPS.

Plan for the next 3–6 months

- 1 **Pilot** with a disability organisation, 2–3 hotels and cultural institutions; an accessibility audit with users.
- 2 **Production sign-in** (transactional e-mail, optional OAuth), notifications, offline mode.
- 3 **AI for checking photos and text:** suggestions to confirm, never facts; the trust engine from Rampa.
- 4 **Open API and a widget** for owners; integration with booking systems.
- 5 **Second city:** city file + OSM extract + city layers; prerequisite: open city data (or OSM alone) and a local partner for moderation.

Not another static map – a living, verifiable picture of a city's accessibility.

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